

Extensions: mixed responses, spatial and temporal ordination

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Outline

- ▶ Mixed response types: modelling communities with different measurement scales
- ▶ Spatial and temporal autocorrelation in latent variables
- ▶ Case study: joint modelling across survey methods with spatial and temporal structure



Not all species are measured the same way

Community data rarely comes in a single format:

- ▶ Species composition: presence-absence
- ▶ Dominant grasses: % cover
- ▶ Rare forbs: count of individuals
- ▶ Invertebrates: biomass

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Standard GLLVMs fit one distribution to all species. But forcing binary, count, and continuous responses into the same family means either:

- ▶ **Information loss** (discretising continuous data)
- ▶ **Model misspecification** (Poisson for proportion data)

Mixed response GLLVMs

In `gllvm`, the `family` argument accepts a **character vector** of length p (one per species column):

```
family_vec <- c(rep("binomial", 12), rep("poisson", 8), rep("
fit <- gllvm(y = Y, family = family_vec, num.lv = 2)
```

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```

Each species gets its own link function and variance structure. The latent variables $\mathbf{u}_i$ are shared across all species.

What distributions are available?

Family	Data type	Link
"gaussian"	Continuous, symmetric	identity
"tweedie"	Non-negative, zero-inflated	log
"orderedBeta"	Proportions including 0/1	logit
"betaH"	Proportions excluding 0 or 1	logit
"poisson"	Counts	log
"negative.binomial"	Overdispersed counts	log
"binomial"	Presence-absence or trials	probit
"ZIP", "ZINB"	Zero-inflated counts	log

Why the latent variable framework makes this possible

The GLLVM is species-specific in its response model:

$$g_j(\mu_{ij}) = \beta_{0j} + \mathbf{x}_i^\top \boldsymbol{\beta}_j + \mathbf{u}_i^\top \boldsymbol{\gamma}_j$$

but the latent variables $\mathbf{u}_i$ are **shared**: they represent the underlying community state at site i .

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This is what makes joint modelling across survey methods possible.

Binomial with $N > 1$ trials

When species are recorded as frequencies (e.g., Raunkiaer: k quadrats occupied out of $N = 100$):

```
fit <- gllvm(y = Y, family = "binomial", Ntrials = 100, num.l
```

`Ntrials` can also be a vector of length p if trials differ per species, or a matrix of size $n \times p$.

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Default link for binomial in `gllvm` is **probit**, not **logit**.

orderedBeta for proportions

Proportions including exact 0 and 1 (e.g., Braun-Blanquet cover converted to $[0, 1]$):

```
fit <- gllvm(y = Y / 100, family = "orderedBeta", num.lv = 2)
```

The orderedBeta distribution places point masses at 0 and 1 with a Beta distribution on $(0, 1)$. More appropriate than "beta" when true absences and full cover both occur.

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zeta.cutoff controls the quantile threshold for the zero/one mixture; default is usually fine, but set explicitly for stability:

```
fit <- gllvm(..., family = "orderedBeta", zeta.cutoff = c(-2,
```

Combining families in a single model

A realistic community dataset might need:

```
family_vec <- ifelse(grepl("_cover$", colnames(Y)), "orderedB",
                    ifelse(grepl("_count$", colnames(Y)), "negative",
                          "binomial"))

Ntrials <- ifelse(grepl("_freq$", colnames(Y)), 100, 1)

fit <- gllvm(y      = Y,
            family = family_vec,
            Ntrials = Ntrials,
            num.lv  = 2,
            zeta.cutoff = c(-2, 2))
```


What does a joint model across datasets look like?

Two surveys, different methods, stacked as rows:

- ▶ Rows $1-n_1$: survey A (ordinal columns only)
- ▶ Rows $n_1 + 1-n_1 + n_2$: survey B (binomial columns only)
- ▶ Shared species appear twice: _COV and _FREQ columns, with NA in the opposite block

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The model estimates **one set of latent variables** covering all rows, and **separate loadings** for _COV and _FREQ versions of each shared species.

If the methods agree, $\hat{\gamma}_{j,\text{COV}} \approx \hat{\gamma}_{j,\text{FREQ}}$; testable with cosine similarity or Procrustes.

A core assumption we have not questioned

All GLLVMs so far assume sites are **independent** after conditioning on the latent variables and covariates.

$$\mathbf{u}_i \stackrel{\text{iid}}{\sim} \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$$

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This fails when:

- ▶ Plots are close in space and share unmeasured environmental variation
- ▶ The same plots are revisited over time
- ▶ The design has nested structure (transects within sites)

Structured covariance for the latent variables

Replace independent LV draws with a **correlated** distribution:

$$\mathbf{u}_{.q} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_q), \quad q = 1, \dots, d$$

where $\boldsymbol{\Sigma}_q$ encodes the dependence structure.

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In `gllvm`: argument `lvCor` specifies the covariance model;
`studyDesign` provides the grouping or ordering variable.

Structures available in glvm

Formula syntax	Structure	Use case
<code>~corAR1(1 time)</code>	AR(1)	Equally-spaced time series
<code>~corExp(1 site)</code>	Exponential kernel	Spatial data (coordinate-based)
<code>~corMatern(1 site)</code>	Matérn kernel	Spatial data (smoother fields)
<code>~corCS(1 group)</code>	Compound symmetry	Nested designs

The same structures are available for row effects via `row.eff`.

You have seen `lvCor` before for nested designs (`corCS`); the same argument accepts spatial and temporal kernels.

Temporal autocorrelation: AR(1)

Community composition at time t is correlated with time $t - 1$:

$$\Sigma_q[t, t'] = \rho^{|t-t'|}, \quad \rho \in (-1, 1)$$

```

fit_ar1 <- gllvm(
  y          = Y,
  X          = Xenv,
  formula    = ~ temperature + precipitation,
  num.lv     = 2,
  studyDesign = data.frame(site = site_id, year = year_id),
  row.eff    = ~corAR1(1|year),
  lvCor      = ~corAR1(1|year),
  family     = "negative.binomial"
)
  
```

Spatial autocorrelation: corExp

Correlation decays exponentially with geographic distance:

$$\Sigma_q[i, k] = \exp\left(-\frac{d_{ik}}{\rho}\right)$$

ρ is the **range parameter**: the distance at which correlation has decayed to $e^{-1} \approx 0.37$.

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Effective range (correlation < 0.05): $d \approx 3\rho$.

```
fit_sp <- gllvm(  
  y           = Y,  
  num.lv      = 2,  
  distLV      = site_coords,    # n x 2 coordinate matrix  
  lvCor       = ~corExp(1|site),  
  studyDesign = data.frame(site = site_id)
```

Spatial vs. temporal: which to use?

Row effects (row.eff)

Community-level variation:
affects all species equally at a
site.

Good for: site productivity,
observer effects, year-level
abundance shifts.

LV correlation (lvCor)

Species composition variation:
different species respond
differently to the latent
gradient.

Good for: spatially or
temporally structured
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Good for: spatially or
temporally structured
community turnover.

Both can be included simultaneously:

```
fit <- gllvm(...,  
  row.eff = ~corAR1(1|year),  
  lvCor    = ~corExp(1|site))
```

The Matérn covariance

A more flexible spatial kernel:

$$K(d) = \sigma^2 \frac{2^{1-\nu}}{\Gamma(\nu)} \left(\frac{\sqrt{2\nu} d}{\rho} \right)^\nu \mathcal{K}_\nu \left(\frac{\sqrt{2\nu} d}{\rho} \right)$$

- ▶ $\nu = 1/2$ recovers the exponential kernel
- ▶ Higher ν produces smoother spatial fields
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Use `corMatern` when you expect smooth spatial gradients rather than abrupt local variation.

Computational considerations

Spatial/temporal GLLVMs require inverting Σ_q , which is expensive for large n .

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gllvm uses **sparse approximations** to the variational covariance matrix:

- ▶ `Lambda.struc = "bdNN"`: block-diagonal nearest-neighbour Cholesky
- ▶ `Lambda.struc = "UNN"`: Kronecker product with NN sparse Cholesky

NN controls the number of neighbours; more neighbours = better approximation, slower fitting.

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For large datasets ($n > 500$), start with NN = 5 and increase if needed.

Concurrent ordination with covariates

Yesterday we saw **concurrent ordination**: LVs constrained to be linear combinations of predictors.

With `num.lv.c` and `lv.formula`, we can combine this with spatial/temporal structure:

```
fit_conc <- gllvm(  
  y           = Y,  
  X           = X,  
  num.lv.c    = 2,  
  lv.formula  = ~ temperature + moisture,  
  distLV      = site_coords,  
  lvCor       = ~corExp(1|site),  
  studyDesign = data.frame(site = site_id),  
  randomB     = "LV"
```

Case study, Garchinger Heide

```
## List of 3
## $ coverage :List of 3
## ..$ Y      : num [1:84, 1:80] 0 0 0 0 0 0 0 0 0 0 0 ...
## ..- attr(*, "dimnames")=List of 2
## ..$ Y_ord: int [1:84, 1:80] 1 1 1 1 1 1 1 1 1 1 1 ...
## ..- attr(*, "dimnames")=List of 2
## ..$ X      : 'data.frame':   84 obs. of  3 variables:
## $ frequency:List of 2
## ..$ Y: num [1:120, 1:116] 0 0 0 0 0 0 0 0 0 0 0 ...
## ..- attr(*, "dimnames")=List of 2
## ..$ X: 'data.frame':   120 obs. of  3 variables:
## $ coords :List of 2
## ..$ coverage : 'data.frame':   42 obs. of  3 variables:
## ..$ frequency: 'data.frame':   40 obs. of  3 variables:
```

Families used in the Garchinger case study

Family	Data type	Link
"ordinal"	Ordered categories (Londo / BB codes)	cumulative probit
"ZNIB"	Zero-inflated count, N trials	log

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`zeta.struc = "common"`: cutpoints shared across species.
 Species-specific cutpoints ("`species`") are unidentifiable under a block-diagonal NA structure.

The Garchinger Heide dataset

A calcareous grassland in Bavaria (Bauer & Albrecht 2020, Basic and Applied Ecology; data: PANGAEA).

Two long-term monitoring programmes, never combined before:

Coverage survey

42 plots $\times$ {2003, 2018}
Londo ordinal codes (12 levels)
→ ordinal

Frequency survey

40 plots $\times$ {1984, 1993, 2018}
Raunkiaer (/100 quadrats)
→ ZNIB, $N = 100$

Question: do the two methods recover the same community gradient?

The joint model design

Stack both surveys as rows; shared species get two columns:

				Bromus_COV	Bromus_FREQ	Molinia_FREQ
COV,	plot	S03,	5		NA	NA
2003						
COV,	plot	S03,	7		NA	NA
2018						
FREQ,	plot	II5,	NA		73	12
1984						
FREQ,	plot	II5,	NA		81	8
2018						

- ▶ 204 rows, 196 columns; 79 shared species $\times$ 2 + survey-only
- ▶ Block-diagonal NA structure: no plot is in both surveys

The identification problem

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The only link is via **shared species**: the model must estimate one loading γ_j that fits both the coverage and frequency observations of species j .

But rotation ambiguity means we cannot guarantee the two blocks are in the same orientation.

We need an external anchor.

Model 1: shared species only

```
fit3 <- gllvm(  
  y          = Y_joint,  
  family     = family_vec,  
  Ntrials    = Ntrials,  
  num.lv     = 2,  
  zeta.struc = "common",  
  row.eff    = ~(1|site),  
  studyDesign = data.frame(site = physical_site)  
)
```

Diagnose alignment: cosine similarity between $\hat{\gamma}_{j,\text{COV}}$ and $\hat{\gamma}_{j,\text{FREQ}}$ for each shared species.

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A bimodal distribution at ± 1 signals rotation ambiguity.

Non-bimodal signals real disagreement.

Model 2: spatial anchor

4 coverage–frequency plot pairs lie within 6 m of each other. The spatial covariance provides a partial anchor via nearby plots.

```
fit4 <- update(fit3,  
  distLV      = coords_sites,  
  lvCor       = ~corExp(1|site),  
  studyDesign = data.frame(site = physical_site),  
  Lambda.struc = "UNN")
```

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```

Lambda.struc = "UNN": sparse nearest-neighbour Cholesky approximation;
see computational considerations slide.

Results: spatial model

Procrustes comparison of coverage and frequency loadings:

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Fitted spatial range: $\hat{\rho}_{LV1} \approx 15$ m, $\hat{\rho}_{LV2} \approx 2481$ m.

Effective range ($\text{Cor} < 0.05$): ~ 44 m and ~ 7432 m.

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Effective range ($\text{Cor} < 0.05$): ~ 44 m and ~ 7432 m.

Most inter-plot distances exceed 80 m; the community gradient is driven by local habitat conditions, not broad spatial autocorrelation. The few close pairs provide just enough anchor.

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- ▶ Erica herbacea: calcicole specialist, sensitive to management changes
- ▶ Molinia caerulea: competitive graminoid, potentially expanding since the 1980s

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Disagreement is concentrated in species that have **changed between the survey periods** (1984–1993 vs. 2003–2018), not random noise.

This is a resurvey: disagreement = ecological change, not methodological failure.

Model 3: adding temporal structure

Year as a concurrent covariate; site random intercepts carry over from fit4:

```

fit5 <- update(fit4,
  num.lv      = 0,
  X           = X_joint,
  num.lv.c    = 2,
  lv.formula  = ~ Year,
  randomB     = "LV",
  lvCor       = ~corExp(1|site),
  Lambda.struc = "UNN")
  
```

What does each component do?

- ▶ `lv.formula = ~ Year`: LVs are driven by survey year; temporal community change is captured in the ordination axes
- ▶ `row.eff = ~(1|site)`: site-level random intercepts absorb overall abundance differences across plots
- ▶ `lvCor = ~corExp(1|site)`: residual spatial structure in community composition after accounting for Year

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- ▶ `lvCor = ~corExp(1|site)`: residual spatial structure in community composition after accounting for Year

The residual loadings alignment then answers: after partialling out temporal change, do the two surveys still agree on the community gradient?

Three model perspectives

Model		Anchor	Alignment	Interpretation	
fit3:	shared species	Shared γ only	Bimodal ± 1 ?	Rotation	ambiguity
fit4:	+ Spatial	Nearby cross-survey pairs	$r \approx 0.2$	Methods	agree spatially
fit5:	+ Year	Space + time	$r \approx 0.04$	Residual	agreement

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The three models are not competitors: they answer **different questions** about the same data.

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The joint family GLLVM makes all three possible within a single framework.

Model comparison

```
AIC(fit3, fit4, fit5)
```

```
##          df      AIC
## fit3 830 65243.53
## fit4 832 72023.04
## fit5 834 66567.71
```

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##          df          AIC
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```

AIC favours the more complex models, but the models target different questions: do not use AIC to choose among them.

Summary

Mixed responses

- ▶ family accepts a vector; each species gets its own distribution and link
- ▶ Shared LVs connect all species into one ordination space
- ▶ Enables joint modelling across datasets with different measurement scales

Spatial and temporal autocorrelation

- ▶ `lvCor` and `row.eff` accept structured covariance models (`corExp`, `corAR1`, ...)
- ▶ `distLV` provides coordinates for spatial kernels
- ▶ Short fitted ranges are informative, not a failure

Combining both